

<https://artificialanalysis.ai/>

<https://www.langchain.com/>

<https://huggingface.co/>

<https://gale.udemy.com/course/complete-agentic-ai-bootcamp-with-langgraph-and-langchain/>

177. Introduction to text summarization With Langchain
https://python.langchain.com/v0.2/docs/tutorials/summarization/
LangChain
Integrations
API Reference
More
v0.2

Introduction
Tutorials
How-to guides
Conceptual guide
Ecosystem
LangSmith
LangGraph
Versions
Overview
Release Policy
Packages
v0.2
Security

Concepts

Concepts we will cover are:

- Using language models.
- Using document loaders, specifically the WebBaseLoader to load content from an HTML webpage.
- Three ways to summarize or otherwise combine documents.
 - Stuff, which simply concatenates documents into a prompt;
 - Map-reduce, which splits documents into batches, summarizes those, and then summarizes the summaries;
 - Refine, which updates a rolling summary by iterating over the documents in a sequence.

That's a fair amount to cover! Let's dive in.

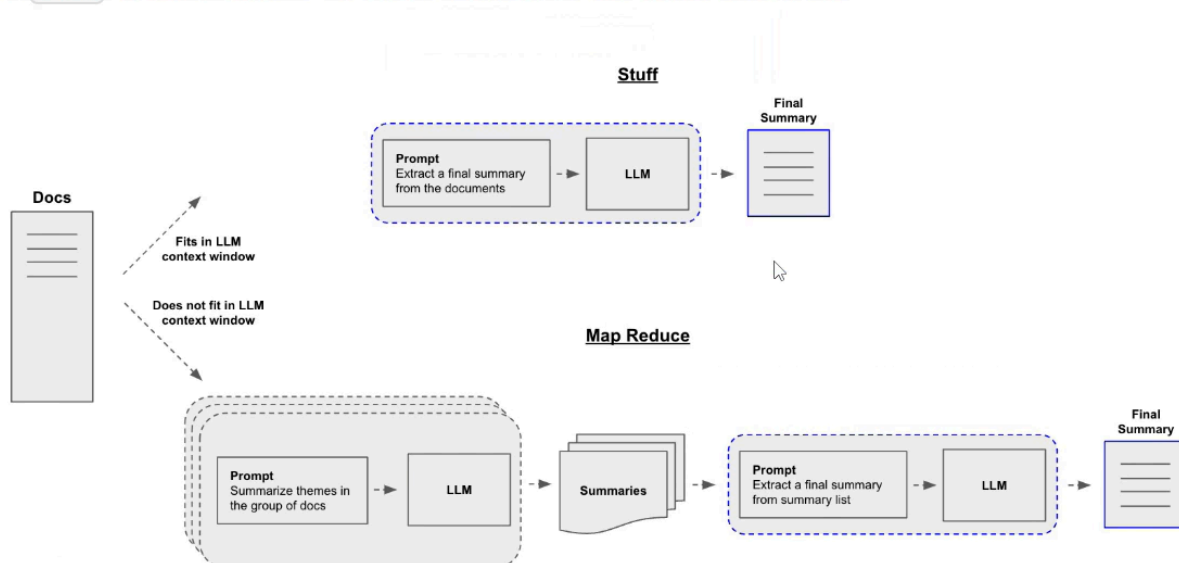
Setup

Jupyter Notebook

This guide (and most of the other guides in the documentation) uses *Jupyter notebooks* and assumes the reader is as well. Jupyter notebooks are perfect for learning how to work with LLM systems because oftentimes things can go wrong (unexpected output, API down, etc) and going through guides in an interactive environment is a great way to better understand them.

Concepts
Setup
Jupyter Notebook
Installation
LangSmith
Overview
Quickstart
Option 1. Stuff
Go deeper
Option 2. Map-Reduce
Go deeper
Option 3. Refine
Splitting and summarizing
Next steps

- Stuff**: Simply "stuff" all your documents into a single prompt. This is the simplest approach (see [here](#) for more on the `create_stuff_documents_chain` constructor, which is used for this method).
- Map-reduce**: Summarize each document on its own in a "map" step and then "reduce" the summaries into a final summary (see [here](#) for more on the `MapReduceDocumentsChain`, which is used for this method).
- Refine**: Update a rolling summary by iterating over the documents in a sequence.



Text Summarization

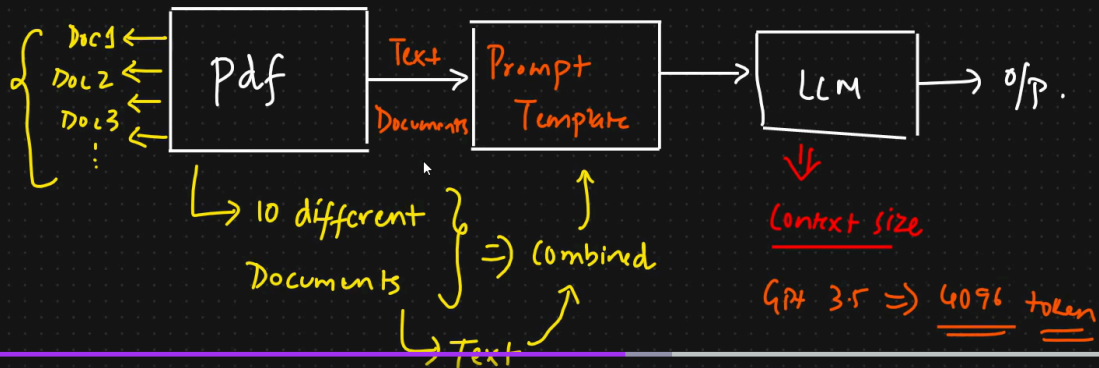
① Stuff Document chain Text Summarization

② Map Reduce Summarization Technique → larger files

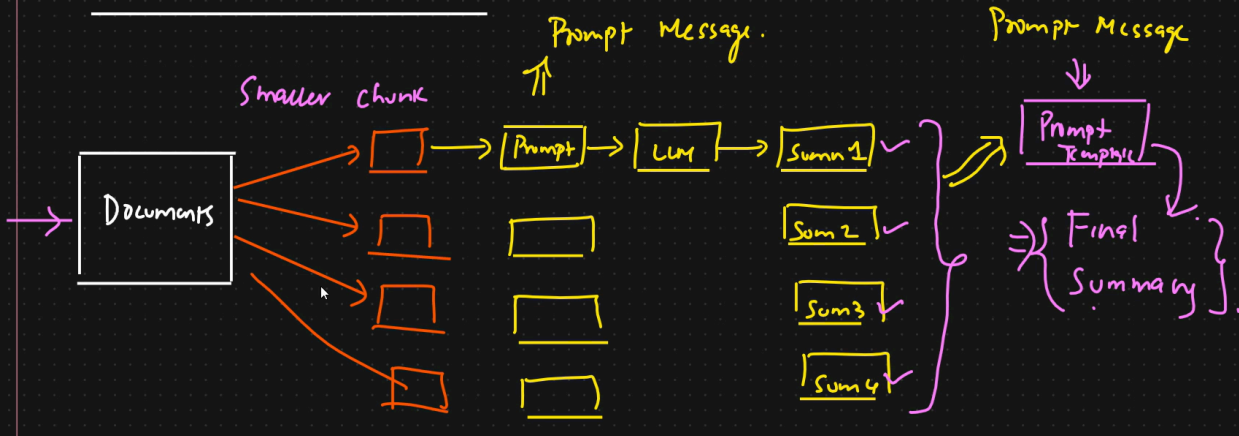
③ Refine chain Summarization

Single Prompt+Template
Multiple Prompt

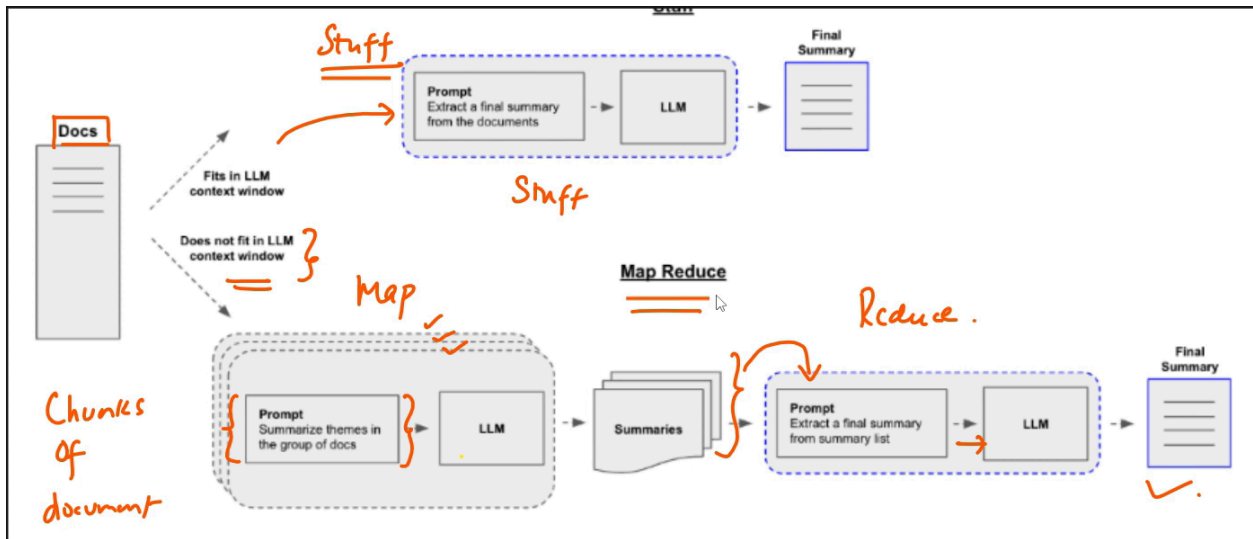
① Stuff Document Chain Summarization



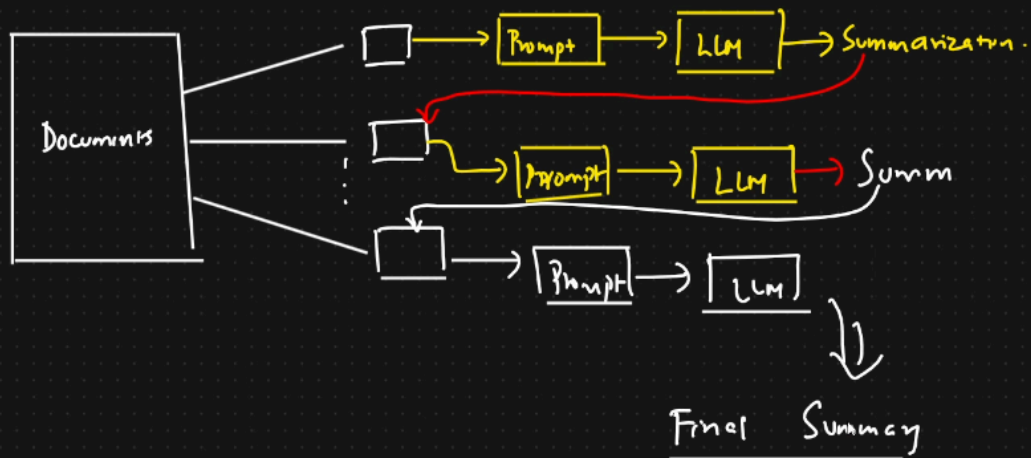
② Map Reduce Summarization



Code is self explanatory



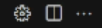
③ Refine chain Summarization





requirements.txt

text_summarization.ipynb



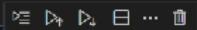
7-Text Summarization > text_summarization.ipynb > ...

+ Code + Markdown | ▶ Run All ↺ Restart 🗑 Clear All Outputs | 📄 Variables 📖 Outline ...

venv (Python 3.10.0)



Refine Chain For Summarization



```
chain=load_summarize_chain(  
    llm=llm,  
    chain_type="refine",  
    verbose=True  
)  
output_summary=chain.run(final_documents)  
output_summary
```

[43]

✓ 13.3s

Python



> Entering new RefineDocumentsChain chain...



> Entering new LLMChain chain...